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Automatic control of phacoemulsification needle trajectory

Abstract

A system and method that includes inserting a needle of a phacoemulsification handpiece into an eye of a patient and vibrating the needle in a first trajectory. Matter from the eye is aspirated via an aspiration line while the needle is vibrated in the first trajectory. An indication is received, of a vacuum level in the aspiration line. Determined is, that the vacuum level has changed by at least a preset vacuum level change, and in response vibrating the needle is switched to a second trajectory, different from the first trajectory.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4954960	12/1989	Lo et al.	N/A	N/A
5157603	12/1991	Scheller et al.	N/A	N/A
5160317	12/1991	Costin	N/A	N/A
5279547	12/1993	Costin	N/A	N/A
5733256	12/1997	Costin	N/A	N/A
5836990	12/1997	Li	N/A	N/A
6690280	12/2003	Citrenbaum et al.	N/A	N/A
7727193	12/2009	Boukhny et al.	N/A	N/A
7842005	12/2009	Kadziauskas et al.	N/A	N/A
8048020	12/2010	Boukhny et al.	N/A	N/A
8246580	12/2011	Hopkins et al.	N/A	N/A
8414605	12/2012	Gordon et al.	N/A	N/A
8523812	12/2012	Boukhny et al.	N/A	N/A
8597228	12/2012	Pyles et al.	N/A	N/A
9144517	12/2014	Kuebler et al.	N/A	N/A
9282989	12/2015	Boukhny et al.	N/A	N/A
9545335	12/2016	Boukhny et al.	N/A	N/A
9707127	12/2016	Kadziauskas	N/A	N/A
9999710	12/2017	Ross et al.	N/A	N/A
10045882	12/2017	Balicki et al.	N/A	N/A
10070988	12/2017	McDonell et al.	N/A	N/A
10219940	12/2018	Raney et al.	N/A	N/A
10368760	12/2018	Hauck	N/A	N/A
10453571	12/2018	Teodorescu	N/A	N/A
10463780	12/2018	Mallough et al.	N/A	N/A
10596033	12/2019	Urich et al.	N/A	N/A
10940039	12/2020	Banko	N/A	N/A

11141313	12/2020	Zhang et al.	N/A	N/A
11317937	12/2021	Nott et al.	N/A	N/A
11399858	12/2021	Sawhney et al.	N/A	N/A
11464559	12/2021	Nott et al.	N/A	N/A
11589888	12/2022	Shelton, IV et al.	N/A	N/A
11612408	12/2022	Yates et al.	N/A	N/A
2004/0092800	12/2003	MacKool	N/A	N/A
2004/0193182	12/2003	Yaguchi et al.	N/A	N/A
2005/0209560	12/2004	Boukhny et al.	N/A	N/A
2005/0261628	12/2004	Boukhny et al.	N/A	N/A
2006/0079788	12/2005	Anderson et al.	N/A	N/A
2006/0100570	12/2005	Urich et al.	N/A	N/A
2006/0129140	12/2005	Todd et al.	N/A	N/A
2007/0073309	12/2006	Kadziauskas et al.	N/A	N/A
2007/0161972	12/2006	Felberg et al.	N/A	N/A
2008/0064935	12/2007	Wong et al.	N/A	N/A
2009/0118663	12/2008	Rockley et al.	N/A	N/A
2009/0182266	12/2008	Gordon et al.	N/A	N/A
2010/0069825	12/2009	Raney	N/A	N/A
2010/0118266	12/2009	Nixon	N/A	N/A
2010/0287127	12/2009	Claus et al.	N/A	N/A
2011/0015563	12/2010	Boukhny	N/A	N/A
2011/0092896	12/2010	Kuebler et al.	N/A	N/A
2011/0144641	12/2010	Dimalanta, Jr. et al.	N/A	N/A
2011/0313280	12/2010	Govari et al.	N/A	N/A
2012/0197215	12/2011	Akahoshi	604/272	A61F 9/00745
2012/0232466	12/2011	Kuebler et al.	N/A	N/A
2013/0211435	12/2012	Boukhny et al.	N/A	N/A
2014/0018724	12/2013	Staggs	N/A	N/A
2014/0114296	12/2013	Woodley et al.	N/A	N/A
2014/0257172	12/2013	Yalamanchili	N/A	N/A
2014/0316254	12/2013	Eversull et al.	N/A	N/A
2014/0323953	12/2013	Sorensen et al.	N/A	N/A
2015/0045806	12/2014	Urich et al.	N/A	N/A
2015/0073816	12/2014	Ha et al.	N/A	N/A
2015/0148615	12/2014	Brennan et al.	N/A	N/A
2015/0216726	12/2014	Kadziauskas et al.	N/A	N/A
2016/0175543	12/2015	Frankhouser et al.	N/A	N/A
2018/0028359	12/2017	Gordon et al.	N/A	N/A
2018/0092555	12/2017	Script	N/A	N/A
2018/0207330	12/2017	Ovchinnikov et al.	N/A	N/A
2018/0318131	12/2017	Boukhny et al.	N/A	N/A
2019/0099526	12/2018	Hajishah et al.	N/A	N/A
2019/0133822	12/2018	Banko	N/A	N/A
2019/0274748	12/2018	Danziger et al.	N/A	N/A
2020/0107957	12/2019	Zhang	N/A	N/A
2020/0107958	12/2019	Wong et al.	N/A	N/A
2020/0309760	12/2019	Durant	N/A	N/A
2021/0045918	12/2020	Raney	N/A	N/A
2021/0196515	12/2020	Urich	N/A	N/A

2021/0330493	12/2020	Steen	N/A	A61F 9/00745
2021/0361481	12/2020	Gliner et al.	N/A	N/A
2021/0401623	12/2020	Zhang et al.	N/A	N/A
2022/0008251	12/2021	Govari et al.	N/A	N/A
2022/0192876	12/2021	Algawi et al.	N/A	N/A
2022/0192878	12/2021	Algawi et al.	N/A	N/A
2022/0313489	12/2021	Hajishah et al.	N/A	N/A
2022/0362452	12/2021	Algawi et al.	N/A	N/A
2023/0039808	12/2022	Govari et al.	N/A	N/A
2023/0285189	12/2022	Govari	N/A	N/A
2023/0320897	12/2022	Hajishah et al.	N/A	N/A
2024/0099884	12/2023	Govari et al.	N/A	N/A
2024/0130891	12/2023	Govari	N/A	N/A
2024/0398618	12/2023	Gliner et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3234621	12/1983	DE	N/A
0741554	12/1995	EP	N/A
0956840	12/1998	EP	N/A
1935383	12/2007	EP	N/A
1765190	12/2007	EP	N/A
1743309	12/2010	EP	N/A
3448235	12/2018	EP	N/A
4126304	12/2007	JP	N/A
9211814	12/1991	WO	N/A
2008016870	12/2007	WO	N/A
2010014937	12/2009	WO	N/A
2016122790	12/2015	WO	N/A
2019069201	12/2018	WO	N/A
2021119616	12/2020	WO	N/A
2023170486	12/2022	WO	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 17/357,587, titled, “Accurate Irrigation Rate Measurment System and Method,” filed Jun. 24, 2021. cited by applicant

U.S. Appl. No. 16/727,100, titled “Phacoemulsification Apparatus,” filed Dec. 26, 2019. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Patent Application 63/230,741, filed Aug. 7, 2021, whose disclosure is incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to phacoemulsification systems, and particularly to control of a phacoemulsification probe tip trajectory.

BACKGROUND OF THE DISCLOSURE

(2) A cataract is a clouding and hardening of the eye's natural lens, a structure which is positioned behind the cornea, iris and pupil. The lens is mostly made up of water and protein and as people age these proteins change and may begin to clump together obscuring portions of the lens. To correct this, a physician may recommend phacoemulsification cataract surgery. In the procedure, the surgeon makes a small incision in the sclera or cornea of the eye. Then a portion of the anterior surface of the lens capsule is removed to gain access to the cataract. The surgeon then uses a phacoemulsification probe, which has an ultrasonic handpiece with a needle. The tip of the needle vibrates at ultrasonic frequency to sculpt and emulsify the cataract while a pump aspirates particles and fluid from the eye through the tip. Aspirated fluids are replaced with irrigation of a balanced salt solution to maintain the anterior chamber of the eye. After removing the cataract with phacoemulsification, the softer outer lens cortex is removed with suction. An intraocular lens (IOL) is then introduced into the empty lens capsule restoring the patient's vision.

(3) The present disclosure will be more fully understood from the following detailed description of the examples thereof, taken together with the drawings in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a pictorial view of a phacoemulsification system constructed to operate in accordance with an example of the present disclosure; and

(2) FIG. 2 is a flow chart that schematically illustrates a method for controlling a phacoemulsification needle, to prevent emulsified particles from sticking, in accordance with an example of the present disclosure.

DETAILED DESCRIPTION OF EXAMPLES

Overview

(3) A phacoemulsification system is used to remove a natural eye lens with a cataract by breaking it into smaller pieces and emulsifying the pieces using a handpiece comprising a hollow needle. The eye lens fragments are then aspirated from the eye through the needle.

(4) During aspiration some emulsified particles may stick to the external wall of the phacoemulsification needle, creating what is termed a “lollipop” effect. Removal of the needle from the eye while particles are stuck to the needle wall may cause trauma to the eye.

(5) Examples of the present disclosure that are described herein provide a method for correcting the possible presence of stuck particles. When the needle first contacts the eye lens, the vacuum level in the line rises (i.e., the pressure decreases). The vacuum level remains high as the needle penetrates into the lens and divides the lens into sections. The vacuum level drops when emulsified material from the sections begins to be aspirated into the aspiration line, and this is a time when emulsified particles may stick to the needle wall.

(6) In one example, a processor looks for the drop in vacuum level described above, and if the drop is detected, the processor toggles the ultrasonic vibration trajectory of the needle within the eye to a different trajectory for a preset period of time, after which the trajectory is switched back to its initial trajectory. The initial trajectory may be longitudinal (with respect to the longitudinal axis of the needle), and the latter trajectory may be elliptical (e.g., a combination of longitudinal and transverse (or side to side) movement with respect to the longitudinal axis of the needle). Toggling in this manner may remove particles stuck to the needle.

(7) More generally, the processor may toggle between a number of different trajectories of

vibrating the phacoemulsification needle, e.g., longitudinal, planar (e.g., transverse), torsional, elliptical, as well as in combinations and sub-combinations of such trajectories.

System Description

(8) FIG. 1 is a pictorial view of a phacoemulsification system 10 constructed to operate in accordance with an example of the present disclosure. FIG. 1 includes an inset 25, and, as shown in the figure and the inset system 10, includes a phacoemulsification probe/handpiece 12 comprising a needle 16. Needle 16 is configured to be inserted into a lens capsule 18 of an eye 20 of a patient 19. Needle 16 is mounted on a horn 14 of probe 12, and is shown in inset 25 as a straight needle. However, any suitable needle may be used with phacoemulsification probe 12, for example, a curved or bent tip needle commercially available from Johnson & Johnson Surgical Vision, Irvine, CA, USA. A physician 15 holds handpiece 12 by a handle 121 so as to perform a phacoemulsification procedure on the eye 20 of patient 19. The physician may activate the handpiece using a foot pedal (not illustrated in FIG. 1).

(9) Handpiece 12 comprises a piezoelectric actuator 22, which is configured to vibrate horn 14 and needle 16 in one or more resonant vibration modes of the combined horn and needle element. During the phacoemulsification procedure, the vibration of needle 16 is used to break a cataract into small pieces.

(10) During the phacoemulsification procedure, an irrigation sub-system 24, which may be located in a console 28, pumps irrigation fluid from an irrigation reservoir to an irrigation sleeve 17 surrounding at least a portion of needle 16, so as to irrigate the eye 20. The combined structure of needle 16 and irrigation sleeve 17 is called hereinafter a “phacoemulsification tip.” The fluid is pumped via a tubing line (irrigation line) 43 running from the console 28 to the probe 12. Irrigation sub-system 24 is described in more detail below.

(11) An aspiration sub-system 26, also typically located in console 28, aspirates eye fluid and waste matter (e.g., emulsified parts of the cataract) from the patient's eye 20 via needle 16 to a collection receptacle (not shown). Aspiration sub-system 26 comprises a pump which produces a vacuum that is connected from the sub-system to probe 12 by an aspiration tubing line 46. A gauge or sensor 47 in line 46 measures the aspiration vacuum and/or pressure. Gauge 47 may be in any convenient location in line 46, including, but not limited to, a location in or in proximity to handpiece 12 or a location in or in proximity to the console 28.

(12) Irrigation sub-system 24 and aspiration sub-system 26 are both controlled by a processor 38. The processor controls the flow volume rate at which the irrigation sub-system pumps fluid. The processor also controls the vacuum, pressure, and/or aspiration rate of the aspiration sub-system, using a pressure reading from gauge 47.

(13) Some or all of the functions of processor 38 may be combined in a single physical component or, alternatively, implemented using multiple physical components. The physical components may comprise hard-wired or programmable devices, or a combination of the two. In some examples, at least some of the functions of processor 38 may be carried out by suitable software stored in a memory 41. The software may be downloaded to a device in electronic form, over a network, for example. Alternatively, or additionally, the software may be stored in tangible, non-transitory computer-readable storage media, such as optical, magnetic, or electronic memory.

(14) Processor 38 may receive user-based commands via a user interface 40, which may include setting and/or adjusting a vibration mode, power level, duty cycle, and/or a frequency of piezoelectric actuator 22, setting and/or adjusting a stroke amplitude of needle 16, and setting and/or adjusting an irrigation rate, an aspiration rate, and/or an aspiration vacuum of irrigation sub-system 24 and aspiration sub-system 26. Additionally, or alternatively, processor 38 may receive user-based commands from controls located in handpiece 12, to, for example, select a trajectory 44, or another trajectory, for needle 16. The implementation of a trajectory such as trajectory 44 is further described below.

(15) Processor 38 may present results of the phacoemulsification procedure on a display 36. In an

example, user interface **40** and display **36** may be one and the same, such as a touch screen graphical user interface.

(16) The system illustrated in FIG. **1** may include further elements, which are omitted for clarity of presentation. For example, physician **15** typically performs the procedure using a stereo-microscope or magnifying glasses, neither of which are shown. Physician **15** may use other surgical tools, in addition to probe **12**, which are also not shown to maintain clarity and simplicity.

(17) Console **28** further comprises a multi-channel piezoelectric drive system **100** comprising drive modules **30.sub.1**, **30.sub.2**, . . . **30.sub.N**, each coupled, using wiring in a cable **33**, to a stack of piezoelectric crystals of piezoelectric actuator **22**. Drive modules **30.sub.1**, **30.sub.2**, . . . **30.sub.N**, generically termed drive modules **30**, are controlled by processor **38** and convey phase-controlled driving signals via cable **33** to piezoelectric actuator **22**. In response, piezoelectric actuator **22** vibrates needle **16**, which performs a vibrational/ultrasound trajectory **44**, the trajectory typically comprising one, or a combination, of longitudinal, transverse, and/or torsional ultrasonic vibrations synchronized one with the other.

Method of Automatic Control of Phacoemulsification Needle Trajectory

(18) As noted above, during the phacoemulsification procedure performed by physician **15**, emulsified matter from the eye **20** of patient **19** is aspirated by aspiration sub-system **26**. However, there are some instances during the procedure when the emulsified matter may stick to the external wall of the needle, causing what is termed a “lollipop” effect. It may be difficult or even impossible for the physician to know that material is stuck to the needle, so that attempts to remove the needle from the eye with the stuck material may cause trauma. Even if the physician is aware of material being stuck to the needle, the only way to remove the material may be to insert another tool, e.g., a chopper or vacuum probe, to collect the material.

(19) FIG. **2** is a flow chart that schematically illustrates a method for controlling a phacoemulsification needle (such as needle **16**), to prevent emulsified particles from sticking, in accordance with an example of the present disclosure. Implementing the steps of the flow chart prevents the lollipop effect from occurring.

(20) During a phacoemulsification procedure, once needle **16** begins to engage with the eye lens, the vacuum level in the aspiration line typically increases due to partial or complete occlusion of the distal end of the needle. During an occlusion the vacuum level may increase to a relatively high level, i.e., the pressure falls to a relatively low level. Typical pressure values for this stage range from approximately -300 mmHg to approximately -650 mmHg. The levels stay generally constant until the needle begins to break the lens into sections.

(21) As the needle breaks up the sections it begins to form emulsified particles, which are aspirated into aspiration line **46**. The aspiration of the emulsified particles at this time reduces the vacuum level in the line, for example, by at least 50 mmHg (i.e., the pressure increases by at least 50 mmHg). This is the time period when some of the particles may stick to the wall of the needle. As is described below, steps of the flow chart check for this reduction in vacuum level, which typically occurs over a relatively short period of time of approximately 3 msec. If this kind of reduction is observed, the flow chart describes actions which may reduce the likelihood of emulsified particles from sticking to the wall of the needle.

(22) In an initial step **180** of the flow chart, physician **15** inserts needle **16** in the lens capsule **18** of patient **19** and provides instructions to processor **38** so that the processor configures drive modules **30** to generate trajectory **44** as a first trajectory, herein assumed to comprise a substantially longitudinal trajectory. Such a longitudinal trajectory may be efficient for initially breaking up the eye lens into small sections.

(23) Processor **38** is also provided with a preset vacuum change level, e.g., 50 mmHg, corresponding to the reduction in vacuum level described above, as well as with a preset period of time, e.g., 3 msec, corresponding to the short time period referred to above. During initial step **180**, processor **38** monitors the pressure in aspiration line **46** using pressure gauge **47**.

(24) In a decision step **184**, while monitoring the pressure in the aspiration line, the processor checks if a vacuum reduction, corresponding to the preset vacuum change level of step **180**, has occurred within the preset time period. If decision step **184** returns negative (“No”), i.e., no vacuum reduction is detected within the preset time period, control returns to step **184**, so that the processor continues to monitor the aspiration line pressure iteratively.

(25) If decision step **184** returns positive (“Yes”), i.e., the processor detects that a vacuum reduction is occurring, control transfers to a switching step **188**.

(26) In switching step **188** of the flow chart, the processor reconfigures drive modules **30** to generate trajectory **44** as a second trajectory, different from the first trajectory, e.g., a substantially elliptical trajectory (combination of longitudinal and transverse). The elliptical trajectory may shake off material stuck to needle **16**. Processor **38** maintains the switch to the second trajectory for a predefined period of time, herein assumed to be a period between approximately 10 μ s and approximately 10 msec.

(27) The example flow chart shown in FIG. **2** is chosen purely for the sake of conceptual clarity. For example, additional steps, such as audiovisual alert may be included.

EXAMPLES

Example 1

(28) A method comprises inserting a needle (**16**) of a phacoemulsification handpiece (**12**) into an eye (**20**) of a patient (**19**), and vibrating the needle in a first trajectory. Matter from the eye (**20**) is aspirated via an aspiration line (**46**) while the needle is vibrated in the first trajectory. An indication is received, of a vacuum level in the aspiration line (**46**). Determined is, that the vacuum level has changed by at least a preset vacuum level change, and in response vibrating the needle is switched to a second trajectory, different from the first trajectory.

Example 2

(29) The method according to example 1, wherein the first trajectory comprises a longitudinal vibration, and the second trajectory comprises an elliptical vibration.

Example 3

(30) The method according to any of examples 1 and 2, wherein switching to vibrate the needle (**16**) in the second trajectory comprises vibrating the needle in the second trajectory for a predefined period of time.

Example 4

(31) The method according to any of examples 1 through 3, further comprising returning to vibrating the needle (**16**) in the first trajectory upon completion of the predefined period of time.

Example 5

(32) The method according to any of examples 1 through 4, wherein determining that the vacuum level has changed comprises determining that the vacuum level has changed by at least the preset vacuum level change during a time period less than or equal to a preset time period.

Example 6

(33) A system (**10**) includes a needle (**16**) of a phacoemulsification handpiece (**12**), an aspiration line (**46**), and a processor (**38**). The needle is configured for insertion into an eye (**20**) of a patient (**19**), and being vibrated in a first trajectory. The aspiration line is configured for aspiration of matter from the eye via the needle while the needle is vibrated in the first trajectory. The processor (**38**) is configured to (i) receive an indication of a vacuum level in the aspiration line, and (ii) determine that the vacuum level has changed by at least a preset vacuum level change, and in response switching to vibrate the needle in a second trajectory, different from the first trajectory.

(34) Although the examples described herein mainly address phacoemulsification, the methods and systems described herein can also be used in other surgical applications that may require a multi-channel piezoelectric resonant system to drive a moving member, such as ultrasonic blades, and other types of actuators.

(35) It will thus be appreciated that the examples described above are cited by way of example, and

that the present disclosure is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present disclosure includes both combinations and sub-combinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art. Documents incorporated by reference in the present patent application are to be considered an integral part of the application except that to the extent any terms are defined in these incorporated documents in a manner that conflicts with the definitions made explicitly or implicitly in the present specification, only the definitions in the present specification should be considered.

Claims

1. A method, comprising: inserting a needle of a phacoemulsification handpiece into an eye of a patient, and vibrating the needle in a first trajectory; aspirating matter from the eye via an aspiration line while the needle is vibrated in the first trajectory; receiving an indication of a vacuum level in the aspiration line; and determining that the vacuum level has changed by at least a preset vacuum level change, and in response switching to vibrate the needle in a second trajectory, different from the first trajectory; wherein determining that the vacuum level has changed by at least a preset vacuum level change comprises determining that the vacuum level has changed by at least the preset vacuum level change during a time period less than or equal to a preset time period.
 2. The method according to claim 1, wherein the first trajectory comprises a longitudinal vibration, and the second trajectory comprises an elliptical vibration.
 3. The method according to claim 1, wherein switching to vibrate the needle in the second trajectory comprises vibrating the needle in the second trajectory for a predefined period of time.
 4. The method according to claim 3, further comprising returning to vibrating the needle in the first trajectory upon completion of the predefined period of time.
 5. A system, comprising: a needle of a phacoemulsification handpiece configured for insertion into an eye of a patient, and being vibrated in a first trajectory; an aspiration line coupled with the needle and configured for aspiration of matter from the eye via the needle while the needle is vibrated in the first trajectory; and a processor, which is configured to: receive an indication of a vacuum level in the aspiration line; and determine that the vacuum level has changed by at least a preset vacuum level change, and in response switching to vibrate the needle in a second trajectory, different from the first trajectory; wherein the processor is configured to determine that the vacuum level has changed by determining that the vacuum level has changed by at least the preset vacuum level change during a time period less than or equal to a preset time period.
 6. The system according to claim 5, wherein the first trajectory comprises a longitudinal vibration, and the second trajectory comprises an elliptical vibration.
 7. The system according to claim 5, wherein the processor is configured to switch to vibrate the needle in the second trajectory by vibrating the needle in the second trajectory for a predefined period of time.
 8. The system according to claim 7, wherein the processor is further configured to return to vibrating the needle in the first trajectory upon completion of the predefined period of time.
 9. The method according to claim 1, wherein the first trajectory is a longitudinal trajectory and the second trajectory is an elliptical trajectory.
 10. The system according to claim 5, wherein the first trajectory is a longitudinal trajectory and the second trajectory is an elliptical trajectory.
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